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HumanErgy Manufacturing - November 2025

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EnMS Logo

ENERGY MANAGEMENT REPORT | ISO 50001:2018

HumanErgy Manufacturing Facility

Energy Performance Report

Reporting Period: **November 2025**

TOTAL ENERGY

275.0 kWh

↓ 3.2% vs previous period

ENERGY COST

$27.5K USD

↑ 1.9% vs previous period

CARBON EMISSIONS

165.0K kg CO_2

↓ 1.8% vs previous period

EFFICIENCY SCORE

87.5 %

↑ 2.5% vs previous period

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CONFIDENTIAL

Executive Dashboard

High-level performance overview and key metrics

TOTAL ENERGY CONSUMPTION

275.0K kWh

▼ 3.2% vs previous month

ENERGY COST

$27.5K

▲ 1.9% vs previous month

CARBON EMISSIONS

165.0K kg CO_2

▼ 1.8% vs previous month

ENERGY EFFICIENCY

87.5 %

▲ 2.5% vs previous month

Top Energy Consumers

Compressor-1	kWh
HVAC-Main	kWh
Conveyor-A	kWh
Pump-Primary	kWh
Furnace-1	kWh

PEAK DEMAND

1,245.3

kW

Occurred at:
N/A

ANOMALIES DETECTED

7

Requires attention

Latest:
Compressor-1

Cost Performance vs Baseline

Expected: **$0.00** | Actual: **$0.00**

OVERSPEND
$0.00
(0%)

Energy Overview

Total consumption analysis for November 2025 (30 days)

TOTAL CONSUMPTION

275.0K

kWh

vs Previous Period ▼ 5.2%

vs Baseline ▼ 3.5%

AVERAGE CONSUMPTION

9.2K

kWh per day

382

kWh per hour

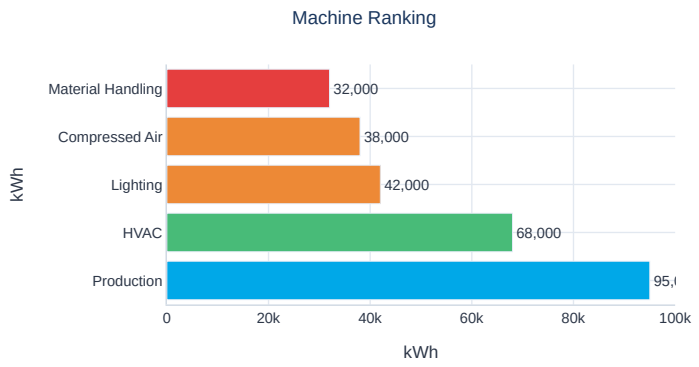
LOAD FACTOR

30.7%

Poor

Peak: 1,245.3 kW
Avg: 381.9 kW

Consumption by Category



Category Rankings

uction — **0 kWh**
hines — 0%

: — **0 kWh**
ines — 0%

ing — **0 kWh**
chines — 0%

pressed Air — **0 kWh**
5 machines — 0%

Material Handling — **0 kWh**
10 machines — 0%

PEAK HOURS

1.2K

kWh average

Peak at: **14:00**

OFF-PEAK HOURS

850

kWh average

Difference: **46.5%**

UTILIZATION

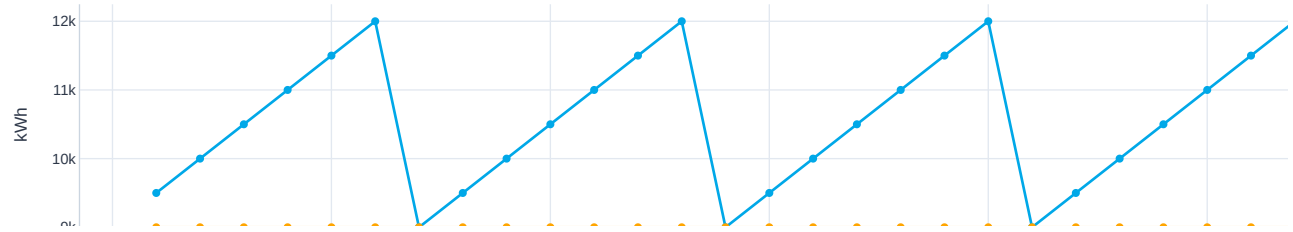
90.3%

of total time

Operating: 650h

Downtime: 70h

Daily Consumption Trend



Hourly Consumption Patterns

Day	00:00	01:00	02:00	03:00	04:00	05:00	06:00	07:00	08:00	09:00	10:00	11:00	12:00	13:00	14:00	15:00	16:00	17:00	18:00	19:00	20:00	21:0
Sun	1008.0	1156.0	905.0	1174.0	1034.0	951.0	1320.0	1085.0	833.0	1261.0	1102.0	1240.0	901.0	874.0	1057.0	1069.0	911.0	1180.0	1212.0	1194.0	1035.0	1357.
Sat	833.0	1267.0	938.0	1304.0	1159.0	1284.0	988.0	1228.0	1211.0	1323.0	1350.0	922.0	1313.0	1351.0	854.0	944.0	1167.0	1390.0	836.0	1106.0	1137.0	1192.
Fri	983.0	1347.0	1376.0	1217.0	1398.0	1108.0	1190.0	876.0	1039.0	1351.0	1309.0	1323.0	871.0	962.0	1024.0	1158.0	1019.0	922.0	823.0	862.0	915.0	935.0
Thu	1324.0	1238.0	962.0	1163.0	1239.0	1136.0	1104.0	1237.0	1204.0	934.0	868.0	935.0	1061.0	1105.0	1150.0	1333.0	1073.0	1286.0	1363.0	1233.0	1105.0	1008.
Wed	1096.0	1376.0	1030.0	1021.0	865.0	839.0	912.0	984.0	980.0	1054.0	1381.0	1169.0	1078.0	970.0	1384.0	1179.0	821.0	992.0	987.0	937.0	1265.0	1218.
Tue	1382.0	824.0	1265.0	942.0	1040.0	1313.0	1387.0	1131.0	973.0	927.0	1024.0	1383.0	1357.0	1078.0	933.0	808.0	1099.0	1090.0	991.0	1140.0	812.0	931.0
Mon	1003.0	813.0	1269.0	922.0	846.0	922.0	1383.0	1047.0	880.0	824.0	874.0	886.0	1255.0	1013.0	1185.0	1217.0	1310.0	1119.0	1314.0	1366.0	1360.0	1338.

Hour

Machine Ranking

Top 5 energy consumers for November 2025

TOP CONSUMER

Compressor-1

Air Compressor

45.2K

kWh (16.4% of total)

MACHINES MONITORED

5

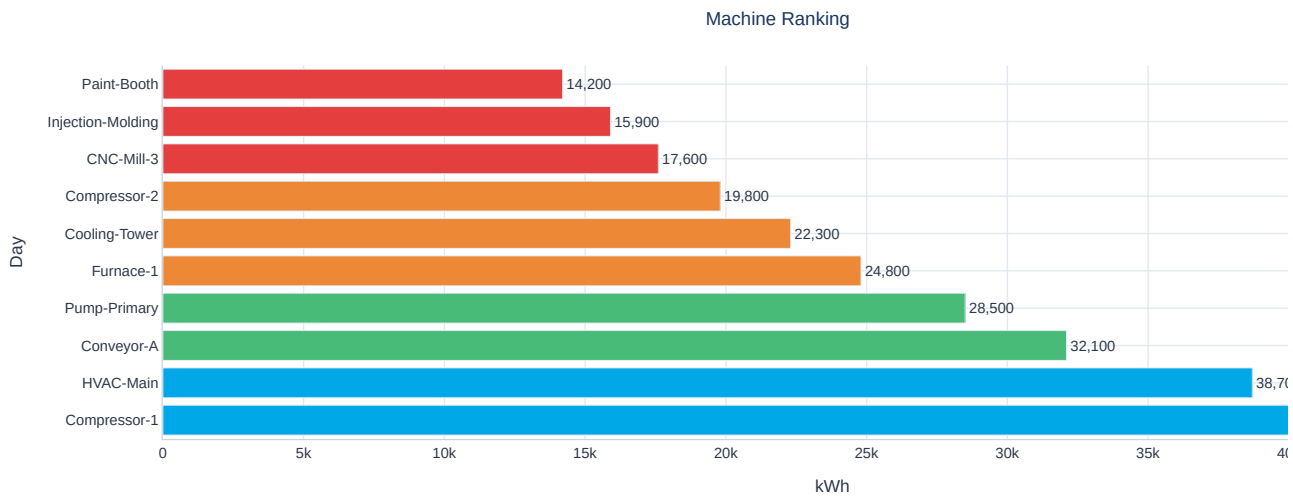
Active equipment units

TOTAL CONSUMPTION

275.0K

kWh total

Consumption Ranking



Detailed Rankings

RANK	MACHINE	TYPE / LOCATION	CONSUMPTION	% OF TOTAL	COST	EFFICIENCY	STATUS
#1	**Compressor-1** 685h operating	Air Compressor Floor A	**45.2K** kWh	16.4%	$4.5K	88.0%	Normal 2 anomalies
#2	**HVAC-Main** 720h operating	Climate Control Central	**38.7K** kWh	14.1%	$3.9K	92.0%	Normal
#3	**Conveyor-A** 650h operating	Material Handling Warehouse	**32.1K** kWh	11.7%	$3.2K	78.0%	Warning 5 anomalies
#4	**Pump-Primary** 700h operating	Water System Utility Room	**28.5K** kWh	10.4%	$2.9K	85.0%	Normal 1 anomalies
#5	**Furnace-1** 580h operating	Heat Treatment Floor B	**24.8K** kWh	9.0%	$2.5K	74.0%	Warning 3 anomalies

Photo

MACHINE PROFILE

Compressor-1

Type: **Air Compressor** Location: **Production Floor A** ID: **COMP-001**

ENERGY CONSUMPTION	EFFICIENCY	ENERGY COST	CO_2 EMISSIONS
45.2K	87.5%	$4.5K	27.1K
↓ 3.5%	↑ 2.3%	Operational expense	kg CO_2

Operating Metrics

Operating Hours **685h**	Utilization **95.1%**
Avg Power **65.9 kW**	Peak Power **73.2 kW**

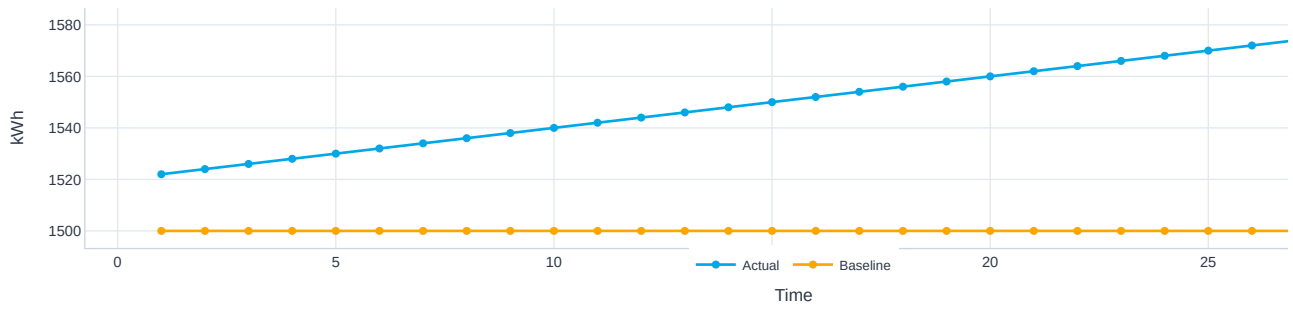
Specific Energy Consumption

Current SEC **0.850 kWh/unit**	Baseline SEC **0.880 kWh/unit**
Production **53,176 m³**	SEC Status GOOD

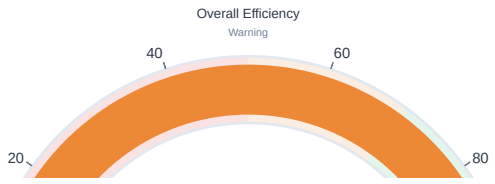
Baseline Comparison

Baseline	Actual	Variance
45.0K	45.2K	+200
		0.4% above

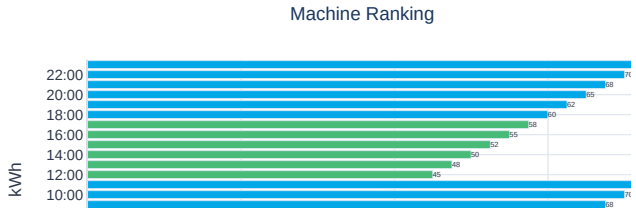
Consumption Trend (Last 30 Days)

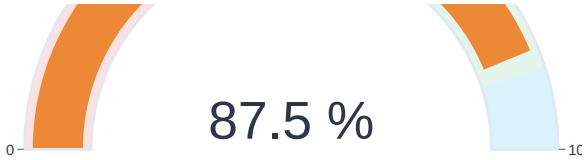


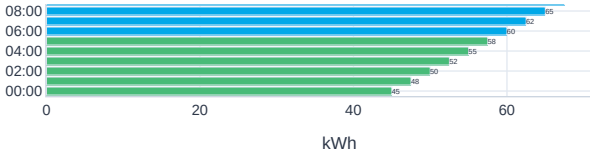
Efficiency Status



Daily Usage Pattern







2 Anomalies Detected

High Power — Severity: High
2025-11-28 14:35
Power exceeded baseline by 25%

Efficiency Drop — Severity: Medium
2025-11-26 09:15
Efficiency dropped to 82%

Recommendations (2)

Schedule Maintenance — HIGH
Recent efficiency drops indicate wear
Potential savings: $850/month

Optimize Schedule — MEDIUM
Shift load to off-peak hours
Potential savings: $420/month

Machine Specifications

Manufacturer	Model	Rated Power
Atlas Copco	**GA 55 VSD+**	**75 kW**

Cost Analysis

Energy cost breakdown and optimization opportunities for November 2025

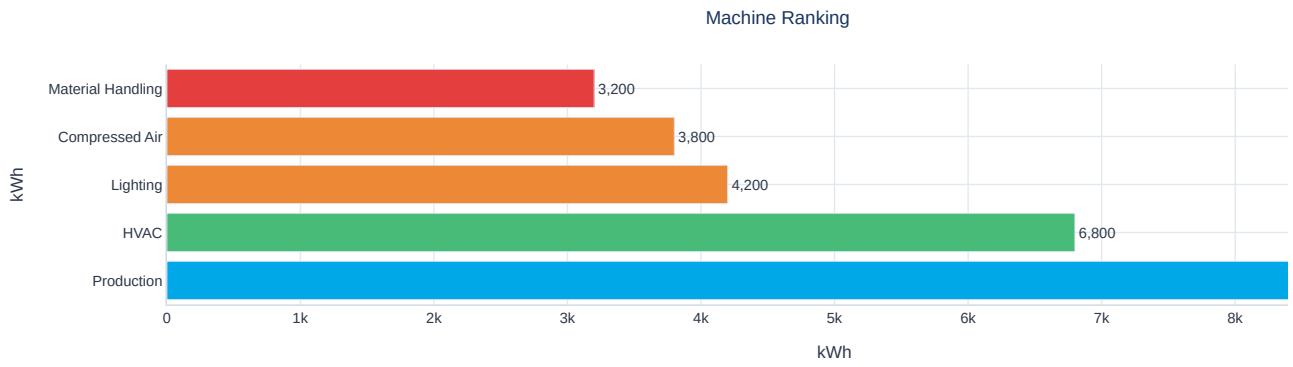
TOTAL ENERGY COST
$27.5K
↑ 1.9% vs previous

BUDGET STATUS
$27.0K
+$500.00 (1.9%)

AVERAGE RATE
$0.100
per kWh

POTENTIAL SAVINGS
$46.8K
170.2% of total

Cost Breakdown by Category



Cost by Category

CATEGORY	CONSUMPTION	COST	% OF TOTAL	AVG RATE
Production	95.0K kWh	**$9.5K**	**34.5%**	$0.1/kWh
HVAC	68.0K kWh	**$6.8K**	**24.7%**	$0.1/kWh
Lighting	42.0K kWh	**$4.2K**	**15.3%**	$0.1/kWh

Peak Hours

Consumption
125.0K kWh

Cost
$12.5K

Rate
$0.135/kWh

45.5% of total cost

Off-Peak Hours

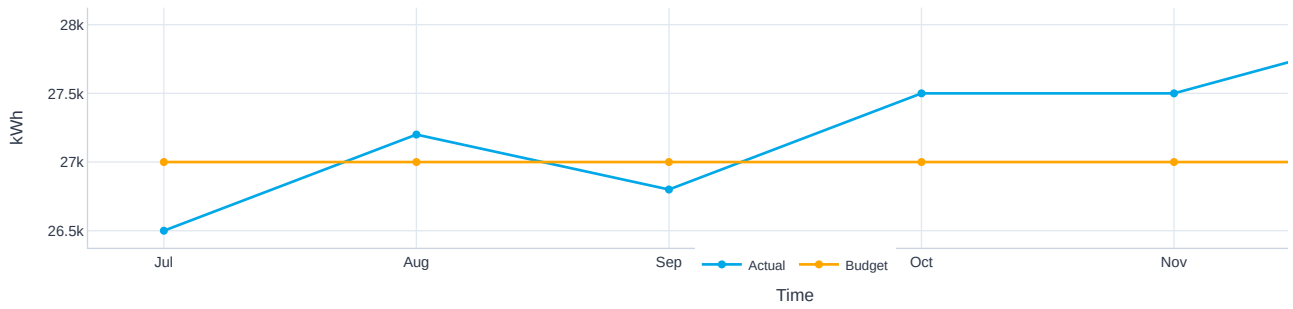
Consumption
85.0K kWh

Cost
$6.8K

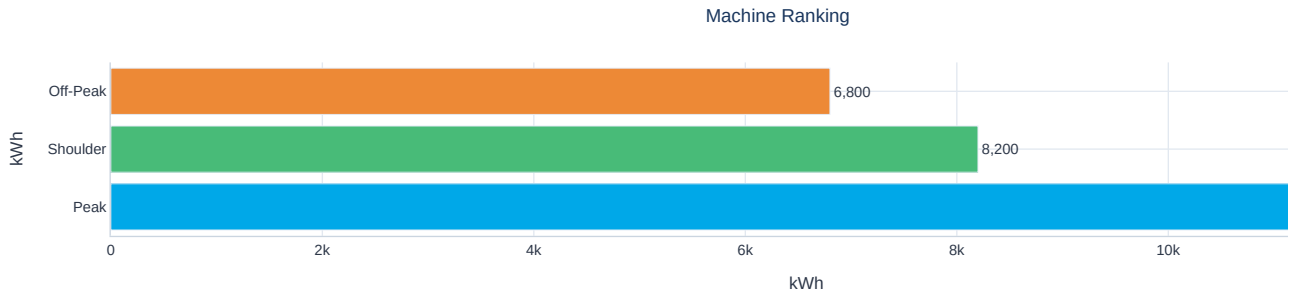
Rate
$0.080/kWh

24.7% of total cost

Cost Trend Over Time



Cost by Tariff Period



Cost Savings Opportunities (1)

Load Shifting HIGH

Shift to off-peak

Savings: $18K **Payback: Immediate**

Annual Projection

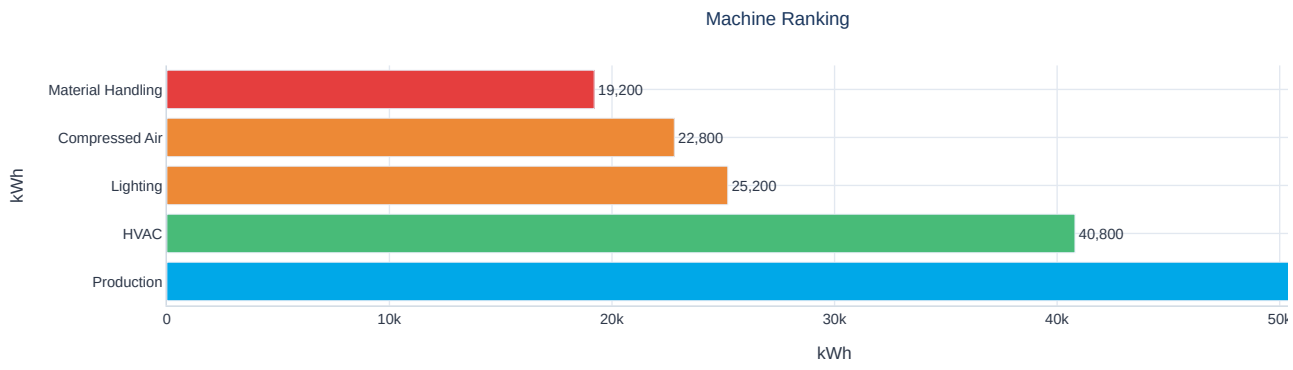
Projected Annual Cost	Potential Annual Savings	Optimized Annual Cost
$330.0K	**$46.8K**	**329438.0**

Carbon Footprint Analysis

Environmental impact and sustainability metrics for November 2025

TOTAL CO_2 EMISSIONS	CARBON INTENSITY	RENEWABLE ENERGY	REDUCTION TARGET
165.0K	0.600	15.0%	24.0%
↓ 1.8% vs previous	kg CO_2/kWh	of energy mix	by December 2030

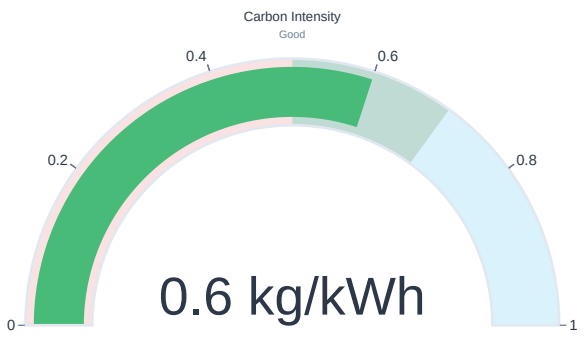
Emissions by Category



Emissions by Category

CATEGORY	ENERGY (KWH)	CO_2 EMISSIONS	% OF TOTAL
Production	95.0K kWh	**57.0K kg**	**34.5%**
HVAC	68.0K kWh	**40.8K kg**	**24.7%**

Carbon Intensity Status

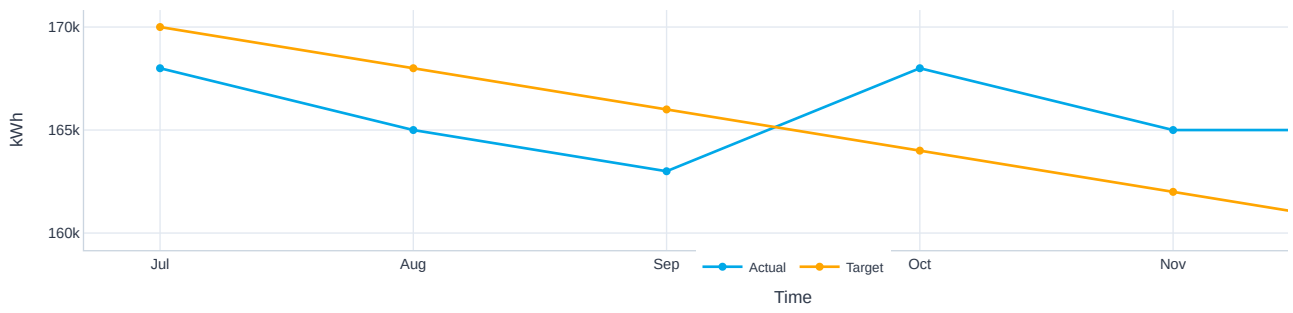


Energy Mix

ıewable
0%
POOR

sil Fuels
0%
Grid Factor: 0.600 kg/kWh

Emissions Trend Over Time



Environmental Context

Trees needed to absorb

15,000

tree-months

Equivalent to driving

410,000

miles in a car

Emission Reduction Initiatives (1)

Solar Installation
250 kW array

IN PROGRESS

Reduction: 75K kg **Timeline: Q2 2026**

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